

Kyle Fernandez

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EDUCATION

York University

Bachelor of Science in Specialized Honours Computer Science

Toronto, ON

Graduation Date: June 2028

EXPERIENCE

Frontend Developer

Freelance – Beyond Education Academy · Seoul, South Korea - Feb 2025 – June 2025

- Developed a **full-stack SAT test simulator** using React (JavaScript) and Tailwind CSS, optimized for both mobile and desktop with responsive layouts and conditional rendering.
- Integrated **Firebase** Authentication and Firestore for user authentication, real-time answer syncing, timer tracking, and session-based test resumption across devices.
- Designed and coded **UI/UX flows** for SAT Reading, Writing, and Math sections, including multi-page navigation, section timers, and score calculation to mimic official testing environments.

AI Automation Developer

Freelance – Paul Academy · Seoul, South Korea - Dec 2024 – Mar 2025

- Developed an AI-driven exam automation tool that **cut exam creation time by 97%**, cutting a 3-hour manual TA process down to under 5 minutes.
- Built a React-based web interface with a JavaScript backend to transcribe and parse PDF exams; automated question extraction and writing into **Google Sheets** using the **Drive & Sheets API**.
- Fine-tuned a **FLAN-T5 model** in Python to classify questions by difficulty, passage type, and question type; supported multi-input and image-based formats.

PROJECTS

Interactive B.MO Bot | Raspberry Pi, Python, Piper-TTS, Vosk, Rhasspy-NLU, Weather-API, TTC API, Google-Calendar API

- Built a **real-time, voice-interactive BMO bot** housed in a custom 3D-printed shell, integrating custom-soldered hardware with a localized AI stack to enable natural, low-latency interaction.
- Integrated **three different Machine Learning** models: **Rhasspy-NLU** for natural language understanding, **Vosk** for offline speech recognition, and **Piper-TTS** for high-fidelity voice synthesis to create a seamless, conversational experience.
- Developed custom automation modules to handle voice-enabled tasks, including fetching live weather updates, calculating **TTC bus arrivals**, and syncing **Google Calendar** schedules for hands-free productivity.

York University Event Hub | Full-Stack Developer | React, Python (FastAPI), Supabase (PostgreSQL), Jira, Trello, Git

- Developed a centralized event platform using **React** and **FastAPI**, integrating a **normalized Supabase (PostgreSQL)** database to provide 50,000+ students with real-time access to campus-wide activities.
- Engineered a **Python-based web scraper** and Discord bot to automate data extraction from club announcement channels, reducing manual entry for organizers and ensuring a dynamic, up-to-date event feed.
- Agile Team Collaboration:** Managed the project lifecycle within a team using **Jira** for sprint planning and **Git** for version control, maintaining code integrity through advanced branching, rebasing, and collaborative conflict resolution.

TECHNICAL SKILLS & CERTIFICATIONS

Languages & Tools: Java, C, Python, JavaScript/TypeScript, Bash, Lua, RISC-V Assembly, SQL, Blender, Figma, Shodan, Wireshark

Frameworks, Libraries & Databases: React.js, Express.js, Three.js, OpenCV, MySQL, Firebase, Supabase, Azure, AWS

Management Platforms: Jira, Trello, Git, Notion, Slack

Certifications & Coursework: CompTIA Security+, Communication Networks / Computer Networks, Software Design, Computer Security

Extracurriculars: Kumon Teacher, Rock Climbing, Music (Piano, Guitar, Banduria)